

The background of the page features a close-up, shallow depth-of-field photograph of pharmaceutical supplies. In the foreground, several white, oval-shaped tablets and two blue-and-white capsules are scattered on a reflective surface. Behind them, two glass vials are visible: one is dark amber with a black cap, and the other is clear with a silver cap and contains a clear liquid. The overall lighting is soft and clinical, with a cool blue-grey color palette.

OLLITAL

TECHNICAL WHITE PAPER

GMP Considerations in Equipment Design

A risk-based guide to hygienic construction, cleaning, containment, controls, qualification and lifecycle documentation.

[/applications/pharmaceutical/](#)

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Decision Brief

GMP expectations become useful equipment requirements only when they are measurable. Product-contact materials, surface finish, drainage, cleanability, containment, controls and documentation must align with the intended product and manufacturing environment.

The supplier supports the evidence package. The manufacturing organization owns intended use, validation strategy, procedures and the final compliance decision.



WHAT THE REVIEW SHOULD DECIDE

01 Begin with a clear URS

State product, process, cleaning, containment, utilities and documentation expectations.

02 Design for inspection

Make product-contact boundaries, access, drainage and routine interventions easy to review.

03 Plan qualification early

Define DQ, FAT, SAT, IQ and OQ evidence before fabrication begins.

GMP Equipment Lifecycle

A traceable lifecycle connects the intended use to design decisions, fabrication records, testing and controlled operation.



Design Feature Review

The useful question is not whether a machine carries a GMP label. The review should identify measurable design features and the evidence that will verify them.



Product-contact materials

Specify grade and compatibility

Request certificates, traceability and a controlled component list. Review corrosion, extractables and mix-up risk.



Surface and drainage

Set inspectable acceptance criteria

Define finish, weld quality, slopes and low points. Plan visual inspection and drain testing.



Seals and interfaces

Make components identifiable

Review compatibility, replacement access, dead spaces and maintenance entry into the product boundary.



Instrumentation

Define criticality and calibration

Set ranges, accuracy, records and calibration access so control evidence remains traceable.

Hygienic Process Interfaces



Hygienic performance depends on the complete product path, including feed, atomization, drying, separation, discharge, cleaning and air handling.

01 Product boundary

Identify every contact surface, gasket, port, sample point and transfer connection.

02 Cleanability

Define manual, COP, CIP or SIP expectations and the evidence required for coverage and drainage.

03 Containment

Review charging, sampling, discharge, filter change and maintenance against the exposure objective.

04 Automation

Connect recipes, access levels, alarms and records to the site's validation strategy.

Risk-to-Design Connections



A quality risk review should connect each credible failure mode to prevention, detection and verification.

Cross-contamination	—	Closed transfer, cleanable interfaces and a documented campaign or changeover strategy.
Product hold-up	—	Drainable geometry, controlled low points and accessible discharge paths.
Uncontrolled intervention	—	Contained sampling, tool access and defined maintenance procedures.
Loss of traceability	—	Material, instrument, software, test and change records maintained through turnover.

Project Input Checklist

The equipment supplier can only design to known requirements. Use the checklist to build a practical URS and assign ownership across the project team.



Product and process

- Product family
- Batch basis
- Critical attributes
- Campaign strategy

Cleaning

- Manual, COP, CIP or SIP
- Cleaning agents
- Residue limits
- Drainage target

Materials

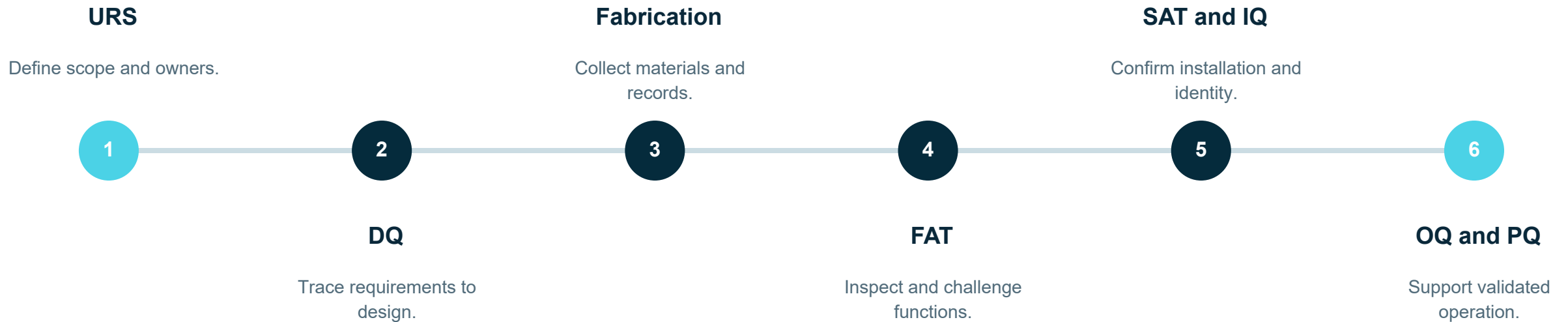
- Contact-metal grade
- Seal expectations
- Surface finish
- Utility contact

Automation and records

- Recipe roles
- Alarm handling
- Data retention
- Qualification scope

Qualification Deliverable Path

Agree the document package before fabrication. Retrofitting evidence after construction is slower and may not close design gaps.



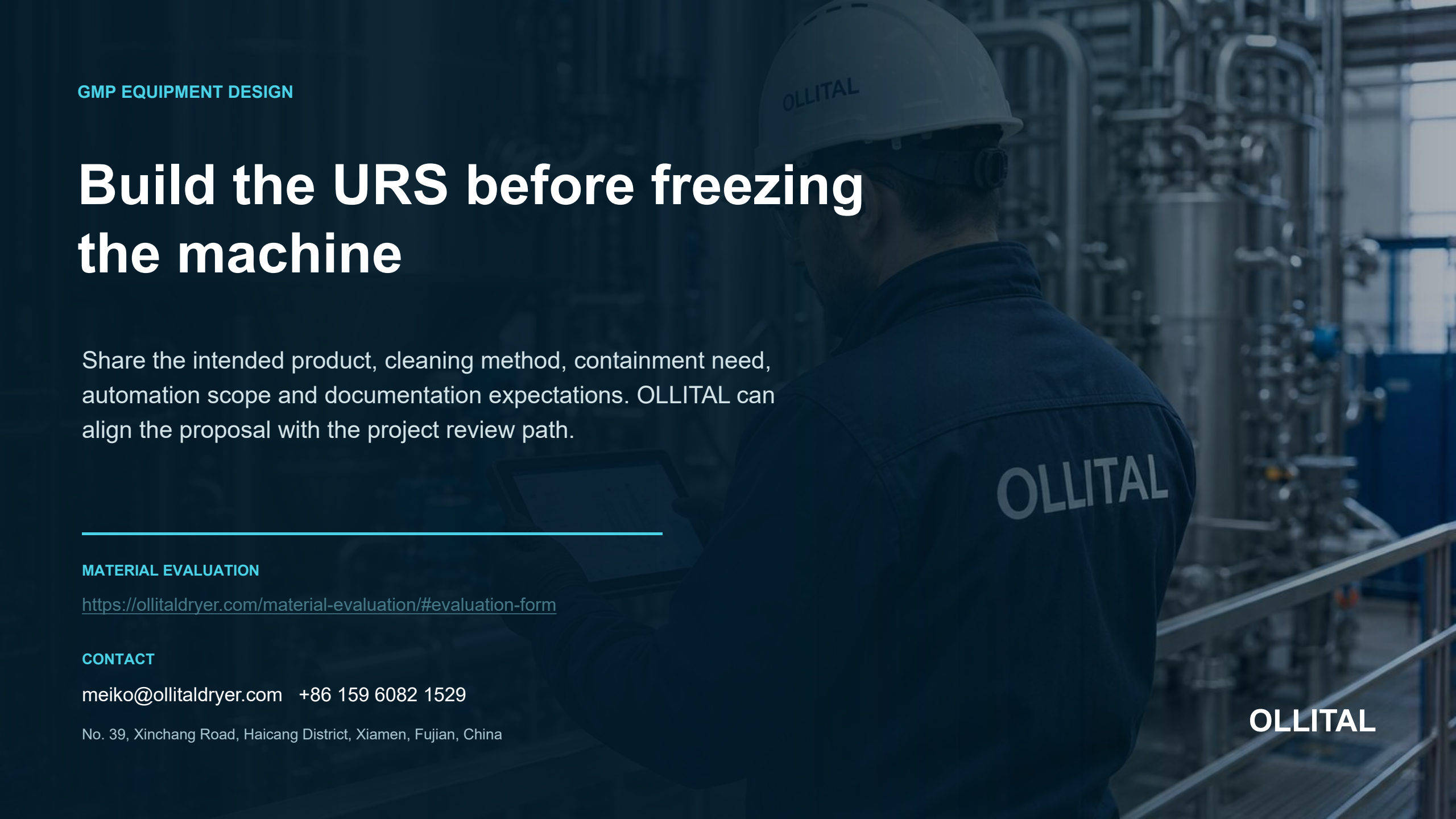
Supplier protocols should support the site's pharmaceutical quality system. The manufacturer determines the final validation approach and acceptance criteria.

Regulatory Review Framework

The applicable GMP basis depends on product type, market and manufacturing route. The quality unit should confirm the final reference set for each project.

01	US FDA 21 CFR 210 and 211	Current Good Manufacturing Practice requirements for drug manufacturing and finished pharmaceuticals.
02	EU GMP Volume 4	Risk-based design, qualification, monitoring and contamination-control expectations.
03	ICH Q7 and ICH Q9	GMP for active pharmaceutical ingredients and principles of quality risk management.
04	Part 11 and Annex 11	Electronic records, electronic signatures and computerized-system lifecycle controls where applicable.





GMP EQUIPMENT DESIGN

Build the URS before freezing the machine

Share the intended product, cleaning method, containment need, automation scope and documentation expectations. OLLITAL can align the proposal with the project review path.

MATERIAL EVALUATION

<https://ollitaldryer.com/material-evaluation/#evaluation-form>

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